

the, is, at

a, an, of



STOP WORDS

in Natural Language Processing

was, were

Filtering out the noise so the signal stands out

"the cat sat on the mat" → ['cat', 'sat', 'mat']

not, it, and

on, to, but

What Are Stop Words?

- Stop words are common words (the, is, at, on, and) that appear very frequently but carry little meaning on their own.
- They're often filtered out before analysis so processing can focus on words that carry the real content.
- There's no single universal list — the right set of stop words depends on the language and the task.

Spot the stop words

The	cat	sat
on	the	mat

 stop word

 content word

“Cat sat mat” carries almost the same meaning as the full sentence — the stop words mostly add grammar, not information.

Why Remove Stop Words?



Noise Reduction

Focuses analysis on the content-bearing words instead of grammatical filler.



Smaller Vocabulary

Shrinks the feature space for classic models like bag-of-words and TF-IDF.



Faster Processing

Fewer tokens to store, index, and compute over across a large corpus.



Better Search Relevance

Search engines skip stop words to match documents on meaningful keywords.

Common English Stop Words

A typical stop word list is dominated by articles, prepositions, conjunctions, and auxiliary verbs:



This is just a sample — typical lists contain 100–300+ words, and vary by library and language.

Example: Removing Stop Words

Input sentence

The quick brown fox jumps over the lazy dog



After stop word removal

quick

brown

fox

jumps

lazy

dog

9 words → 6 tokens. “The,” “over,” and “the” drop out, while “quick,” “brown,” “fox,” “jumps,” “lazy,” and “dog” remain — the words that actually describe what's happening.

Example: Stop Words in Search

Search query

"What is the capital of France?"

MATCHED ON EVERY WORD

what

is

the

capital

of

france

Matches any page that happens to contain "is" or "of" — too broad to be useful.

MATCHED ON KEY TERMS ONLY

capital

france

Matches documents specifically about the capital of France — sharper, more relevant results.

WORKED EXAMPLE

Example: Vocabulary Reduction

Removing stop words shrinks the unique-word vocabulary a bag-of-words model has to track.

Sample corpus: 3 short product reviews

```
"This is the best laptop that I have used."  
"The battery life of this laptop is great."  
"I would not buy this again."
```

Unique words (vocabulary size)

Before removal



24 words

After removal



14 words

Roughly 40% fewer unique tokens to index — in larger corpora the effect is even bigger.

When NOT to Remove Stop Words



Sentiment Analysis

"not good" → "good" — removing "not" flips the meaning entirely.



Phrases & Idioms

"to be or not to be" — here the stop words carry the whole meaning.



Machine Translation

Word order and function words matter for grammatically correct output.



Question Answering

"who", "what", "where" are often stop words — but they carry the question's intent.

Custom Stop Word Lists

General-purpose lists miss words that are frequent — and meaningless — only within a specific domain.

General English stop words

the

is

at

of

and

a

to

+ News-domain additions

said

according

reuters

reported

today

In a corpus of news articles, “said” and “according” appear in almost every document — just like “the” does. Adding them to the stop list keeps the vocabulary focused on what each article is actually about.

Stop Words Across Languages

Language	Example stop words
English	the, is, at, on, and
Spanish	el, la, de, que, y
French	le, la, de, et, un
German	der, die, und, ist, zu
Hindi	और, है, को, से, का

There is no universal stop word list — each language (and often each library) defines its own, so always check what a tool actually filters before you rely on it.

Python Quick Reference

```
nltk.corpus.stopwords.words('english')
```

Returns NLTK's built-in English stop word list

```
token.is_stop
```

spaCy flags each token as a stop word automatically

```
sklearn ENGLISH_STOP_WORDS
```

Built-in list used by scikit-learn's text vectorizers

```
>>> from nltk.corpus import stopwords
>>> stop_words = set(stopwords.words('english'))
>>> tokens = ["the", "cat", "sat", "on", "the", "mat"]
>>> [w for w in tokens if w not in stop_words]
['cat', 'sat', 'mat']
```

Stop Word Removal — Step by Step

Step 1 — Raw text

"The cat sat on the mat"

Step 2 — Tokenize

The

cat

sat

on

the

mat

Step 3 — Lowercase

the

cat

sat

on

the

mat

Step 4 — Remove stop words

cat

sat

mat

By Step 3, “the” and “on” are recognizable stop words (faded); Step 4 removes them, leaving only the content words.

Common Pitfalls

Pitfall	What happens	Fix
Removing negations	"not happy" → "happy" (meaning flipped)	Keep negation words for sentiment tasks
Over-aggressive custom lists	Domain-relevant words get filtered by accident	Review the list against the task's goals
Case-sensitivity bugs	"The" not matched by a lowercase list ["the"]	Lowercase text before filtering
One-size-fits-all lists	Using an English list on non-English text	Use a language-specific stop word list

Key Takeaways

1

Stop words are frequent, low-information words (the, is, at) usually filtered out before analysis.

2

Removing them shrinks vocabulary and speeds up classic techniques like bag-of-words and search indexing.

3

But context matters — sentiment analysis, translation, and question answering often need to keep words like “not” and “who.”

4

Lists are language- and domain-specific — always tailor them, or skip removal entirely, based on the task.

Tip: before removing stop words, ask whether your task needs grammar and negation — not just keywords.



Thank You

Questions about stop words in NLP?